25901386 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCMAT202 - APPLIED MATHEMATICAL METHODS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type Questions

Answer any **5** questions

Each question carries **2** marks

1. Find roots of the equation $x^2 - 4x - 12 = 0$. [U] / [CO1]
2. In the Regula Falsi method, if x is the exact root of the equation, what is the value of $f(x)$? [U] / [CO1]
3. Give an example of a non-periodic function. [A] / [CO2]
4. Define an odd function. Check whether the function $f(x) = x + 1$ is odd or not. [A] / [CO2]
5. Find $L^{-1}(\frac{1}{s+a})$. [U] / [CO3]
6. Find the Laplace transform for $\int_0^t e^{-2t} dt$. [U] / [CO3]
7. Find the integral curves of the equations $\frac{dx}{1} = \frac{dy}{-1} = \frac{dz}{1}$. [U] / [CO4]

8. What do you mean by general solution of Lagrange's equation? [A] / [CO4]

[2x5 = 10]

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **6** marks

9. Solve $f(x) = x^3 - 5x + 2 = 0$ using the iteration method (two iterations). [A] / [CO1]

10. Write the Newton–Raphson iteration formula for $e^x - 3x = 0$ and compute the first three approximations, taking $x_0 = 0.5$. [U] / [CO1]

11. Find the coefficient a_n in the Fourier series expansion of the periodic function $f(x) = k, -\pi < x < \pi$. [A] / [CO2]

12. Find the Fourier series of the function f(x) with period 2π, where $f(x) = 1, -\pi < x < \pi$. [A] / [CO2]

13. Let$f(t) = t\sin(\omega t)$ find $\mathcal{L}\{f''(t)\}$ and $\mathcal{L}\{f(t)\}$. [A] / [CO3]

14. Find the Laplace transform of the function $f(t) = 3.8te^{2.4t}$. [A] / [CO3]

15. Find the integral curves of the equations $\frac{dx}{y^2} = \frac{dy}{-xy} = \frac{dz}{x(z-2y)}$. [E] / [CO4]

16. Solve $(y-1)p + (x-1)q = z$. [A] / [CO4]

[6x5 = 30]

Part C

Essay Type Questions

Answer any **3** questions

Each question carries **10** marks

17. Use the Bisection Method to find the positive root of $f(x) = x^3 - 2x - 5 = 0$ in the interval $[2, 3]$. Perform five iterations. [U] / [CO1]

18. Find the Fourier series of the periodic function [A] / [CO2]

$$f(x) = \begin{cases} x, & 0 < x < 1, \\ 1 - x, & 1 < x < 2, \end{cases} \qquad p = 2L = 2, \quad L = 1.$$

19. Sketch the periodic extension of the function $f(x)$ defined by $f(x) = 3x^2, \ (-1 < x < 1)$ to the entire real line with period 2. Find the Fourier series of $f(x)$. [A] / [CO2]

20. Find the Laplace transform of the functions [A] / [CO3]
    (a) $f(t) = \cos \omega t$
    (b) $f(t) = \sin \omega t$

21. [U] / [CO3]
    1. Find the Laplace transform of the function $f(t) = te^{-}t$.
    2. Find $L(f)$ if $f(t) = 3te^{t}$.

22. Form partial differential equations by [A] / [CO4]
    a) Eliminating the arbitrary function f from $z = f(\frac{xy}{z})$.
    b) Eliminating the constants a and b from $(x-a)^2 + (y-b)^2 + z^2 = 1$.

[10x3 = 30]